

Write your questions and thoughts here!

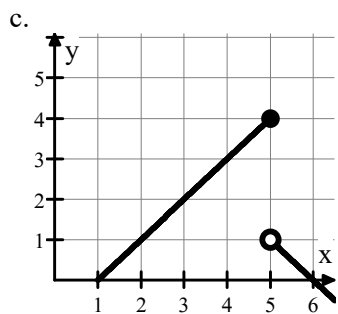
1. If f is a piecewise function with two linear “pieces” such that $\lim_{x \rightarrow 5} f(x)$ does not exist, which of the following could be representative of the function f .

a.

$$f(x) = \begin{cases} 2x - 1, & x < 5 \\ 14 - x, & x > 5 \end{cases}$$

b.

x	2	3	4	5	6	7	8
$f(x)$	10	8	6	1	6	11	16



2. Find the value of $\lim_{x \rightarrow 4^-} \frac{|x-4|}{x-4}$

1.9 Connecting Multiple Representations of Limits

Calculus

Evaluate each limit.

1. $\lim_{x \rightarrow 7^-} \frac{ x-7 }{x-7}$	2. $\lim_{x \rightarrow 3^+} \frac{x-3}{ x-3 }$	3. $\lim_{x \rightarrow -1^+} \frac{ x+1 }{x+1}$
4. $\lim_{x \rightarrow 8^+} \frac{ x-8 }{x-8}$	5. $\lim_{x \rightarrow 2^-} \frac{ x-2 }{x-2}$	6. $\lim_{x \rightarrow -9^-} \frac{x+9}{ x+9 }$

7. Let f be a function where $\lim_{x \rightarrow 5} f(x) = \frac{1}{4}$. Which of the following could represent the function f ?

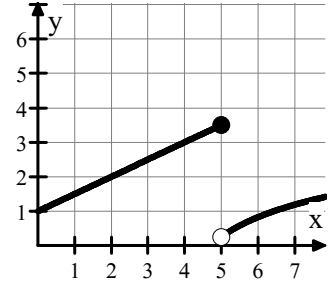
a.

$$f(x) = \begin{cases} \frac{x-5}{x^2-6x+5}, & x \neq 5 \\ 4, & x = 5 \end{cases}$$

b. The two “pieces” are linear.

x	4.8	4.9	4.999	5	5.001	5.1	5.2
$f(x)$	0	0.2	0.249	4	0.251	0.3	0.5

c.



8. Let g be a function where $\lim_{x \rightarrow 3} g(x) = 6$. Which of the following could represent the function g ?

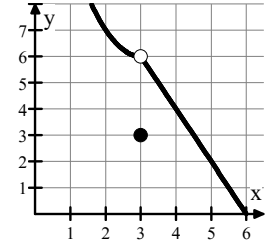
a.

$$g(x) = \begin{cases} \frac{x^2-x-6}{x-3}, & x \neq 3 \\ 6, & x = 3 \end{cases}$$

b. The two “pieces” are linear.

x	2.8	2.9	2.999	3	3.001	3.1	3.2
$g(x)$	6.2	6.01	6.001	1	4.999	4.9	4.8

c.



9. If h is a piecewise function with two linear “pieces” linear function such that $\lim_{x \rightarrow 4} h(x)$ does not exist, which of the following could be representative of the function h ?

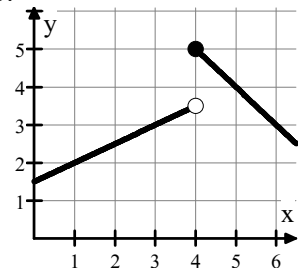
a.

$$h(x) = \begin{cases} \frac{1}{2}x + 3, & x < 4 \\ 13 - 2x, & x > 4 \end{cases}$$

b.

x	1	2	3	4	5	6	7
$h(x)$	-4	-1	2	1	$\frac{16}{3}$	$\frac{17}{3}$	$\frac{18}{3}$

c.



10. If $f(x) = \begin{cases} \frac{(x-3)^2(x^2+1)}{|x-3|} & \text{for } x \neq 3 \\ 2 & \text{for } x = 3 \end{cases}$, then $\lim_{x \rightarrow 3} f(x)$ is

(A) 0

(B) 2

(C) 10

(D) Nonexistent